The Lancet Countdown on Health and Climate Change

Policy brief for China

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LANCET COUNTDOWN: TRACKING PROGRESS ON HEALTH AND CLIMATE CHANGE

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Introduction

In 2020, the COVID-19 pandemic has been not only a crisis of public health, but has also severely affected the global economy, underscoring the critical nature of public health as a prerequisite for economic development. Mounting evidence shows that climate change increasingly poses a wide range of threats to human health in China, and integrated planning with attention to climate, health and economic priorities is crucial. In fact, climate, health, and economic objectives are not only mutually dependent but mutually reinforcing. Multiple case studies have demonstrated that early and ambitious climate mitigation measures offer tremendous health co-benefits and economic savings.

Following China's pledges to reach peak CO_2 emissions before 2030 and achieve carbon neutrality by 2060, designing and implementing a 'triple win' COVID-19 recovery plan and an updated Nationally Determined Contribution to the Paris Agreement could together preserve the climate, protect public health, and promote economic sustainability.

This policy brief presents data from both the 2020 global and China Lancet Countdown reports across four areas: the health and economic impacts of heat; climate-sensitive infectious diseases; detection, preparedness and response to health emergencies; and investment in a low-carbon economy.

Key messages and recommendations

1. **Overarching recommendation: Implement a *'triple win'* COVID-19 recovery plan and an updated Nationally Determined Contribution (NDC) together with increasingly ambitious long-term greenhouse gas emission reduction strategies in line with China's carbon neutrality pledge which together preserve the climate, protect public health, and promote economic sustainability.** While these policies will be finalised over the course of the coming months, they will define societies for decades. Climate, health, and economic objectives are not only mutually dependent but mutually reinforcing.

2. **Formulate a proactive, multi-sectoral policy response to minimise the acute effects of extreme heat on human health.** This process should include collaboration between the energy, transport, building and healthcare sectors, with particular attention to the health of heat-vulnerable populations including elderly people and those with pre-existing conditions. Specific measures are also required to protect workers against heat stress, such as implementation of heat stress prevention programs, training workers in heat stress awareness, increasing the air circulation in the workplace, and ensuring availability of drinking water.

3. **Strengthen monitoring systems to advance understanding of how climate influences vector-borne disease transmission and to enable adequate planning and prevention measures to be implemented to reduce the burden on health systems.** Establishment of such surveillance systems will require collaboration between public health and meteorological agencies to exchange relevant data.

4. **All provinces are encouraged to strengthen their management capacities to respond to health emergencies, especially those related to climate change.** This will require accompanying financial resources. In addition, measures should be taken to improve monitoring, prevention, mitigation, preparedness, response and recovery. Special attention should be given to developing proactive approaches to enhance planned and coordinated rapid-response capability to public health threats.

5. **Boost renewable energy investment in economic stimulus plans and beyond.** This will ensure a sustainable and healthy economic recovery from COVID-19, and keep China on track to meet the goals of the Paris Agreement and Sustainable Development Goals.

The health and economic impacts of heat

Heat and heatwaves pose especially dangerous threats to elderly populations, in part due to the elevated prevalence of chronic diseases such as cardiovascular disease, diabetes, chronic kidney disease and respiratory disease in these age groups.

Similar to the global Lancet Countdown report,[1] the 2020 Chinese Lancet Countdown report tracks the number of days of heatwave exposure experienced by people aged over 65 from 2000 to 2019 compared to a 1986-2005 baseline, using a heatwave definition more sensitive to the Chinese population.[2] Here, a heatwave is defined as a period of three or more consecutive days where the daily maximum temperature was greater than the 92.5th percentile of the distribution of the 1986-2005 baseline daily maximum temperature, a definition found to best capture the health effects of heatwave events in China.[3] The data indicates that heatwave exposure surged from 0.22 billion additional days in 2000 to a record high of 3.54 billion days in 2019, the equivalent of almost every person aged over 65 enduring 21 additional days of heatwave (Figure 1). The three provinces/administrative regions with the highest heatwave exposure per person aged over 65 in 2019 were Hong Kong (49 days), Hainan (31 days) and Yunnan (26 days).[2] The excess risk of death for elderly people has been estimated to increase by over 10% during heatwave days. [4] Thus the rapid increase in the heatwave days endured per person presents a clear threat to national public health.

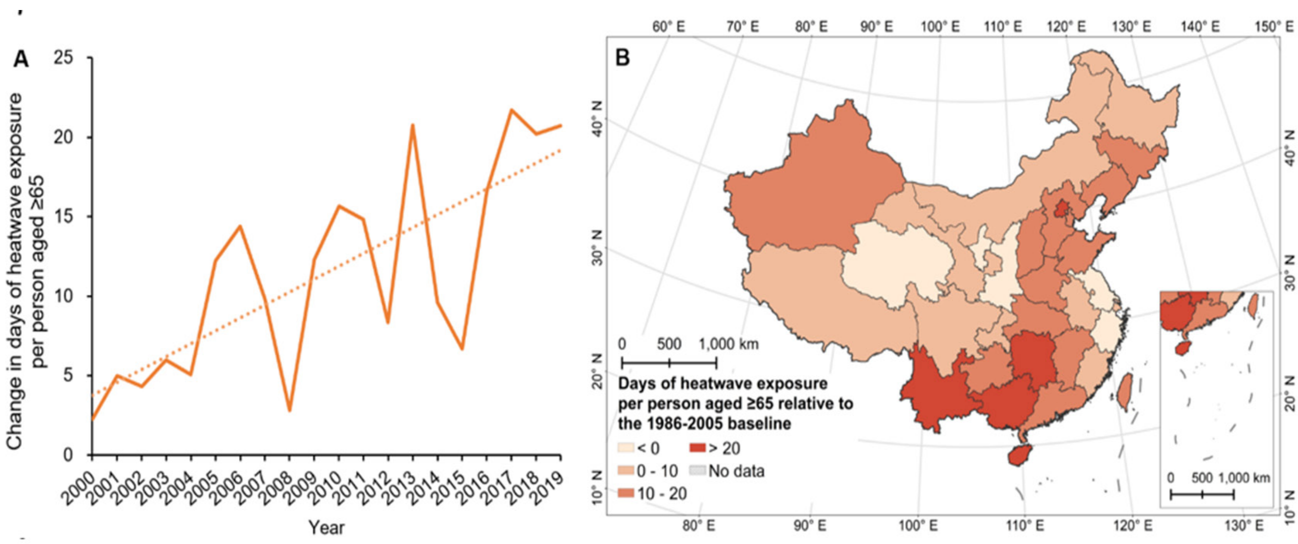

Figure 1: Change in the number of heatwave exposure for each person ≥65 in China, relative to the 1986-2005 average. (A) Country-level results; (B) Provincial-level results in 2019.

Heat stress also impacts the working population, reducing labour productivity, with extensive economic consequences.[5,6,7] In 2019, the potential total work hours lost were over 9.9 billion in China – 4.8% higher than in 2000 and representing over 0.5% of total national work hours.[2] Heat-related labour productivity losses result in substantial economic costs, which consist of both direct (resulting from first-order losses of labour capacity in a particular industry) and indirect losses (in dependent downstream industries). Absolute economic costs of labour productivity loss in 2017 were RMB785 billion (1.14% of GDP), nearly four times the costs arising in 2007 and equivalent to the scale of national fiscal expenditure on science and technology (Figure 2 A).[2] The regional distribution of economic costs clearly reflects China's geographical climate patterns, with southern central China suffering costs equivalent to a higher proportion of regional GDP than other parts of the country (Figure 2B). Nearly a quarter (2.4 billion) of these potential work hours losses were seen in Guangdong, the most populous and economically-developed province, accounting for 11% of China's GDP.[2] The economic costs in Guangdong reached 1.65% of regional GDP in 2015, the highest among all provinces, followed by Hainan (1.41%) and Guangxi (1.22%).[2]

The frequency and severity of heat extremes will systematically increase with future global warming,[8] accompanied by rising heat-related mortality and labor productivity loss. The current rate of growth in health professionals and hospital beds available per capita (5-9%) is far outpaced by the rising incidence of heat-related maladies. Without intervention, climate change will lead to soaring healthcare demand, unmatched by healthcare system capacity, and intensifying healthcare resource competition. Widespread use of air conditioning, as China has witnessed during the past decade, offers an effective adaptation strategy to reduce the health impacts of rising temperature, but associated chlorofluorocarbon and CO_2 emissions from fossil-fuel derived electricity drive climate change. Therefore, measures including a proactive, multi-sectoral policy response which includes increased levels of renewable electricity, implementation of green building standards to reduce the buildings' ecological footprint and improve thermal comfort; and promotion of vehicle fleet electrification and green infrastructure construction to reduce GHG emissions and prevent the urban heat island effect, is essential to minimise the acute effects of extreme heat on vulnerable populations, and avoid economic losses from heat stress at work.

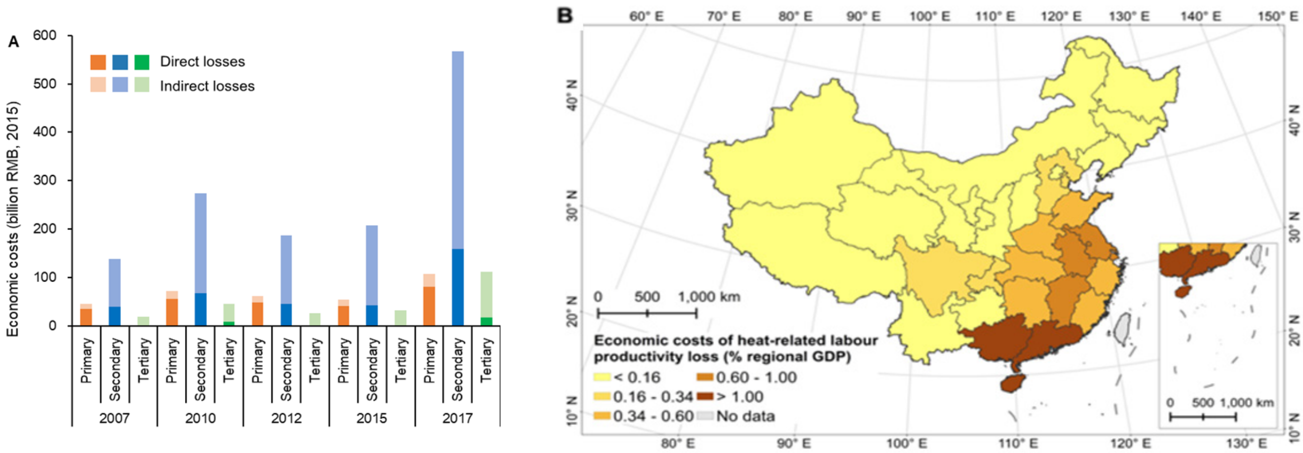

Figure 2: Economic costs of heat-related labour productivity loss, (A) National-level results, by year and industry, in billions of 2015 RMB (B) Provincial-level results in 2015, relative to GDP

Climate-sensitive infectious diseases

Climate suitability for the transmission of vector-borne disease is rising in China. Both the 2020 global and Chinese Lancet Countdown reports track the change in vectorial capacity (VC) of the ***Aedes aegypti (A. aegypti)*** and ***Aedes albopictus (A. albopictus)*** mosquitos to transmit dengue, which is influenced by daily temperature and expressed as the average number of daily cases resulting from one infected case. In China, the climate suitability for the transmission of dengue in 2014-2015 has risen by 37% and 14% since the early 1960s for ***A. aegypti*** and ***A. albopictus***, respectively (Figure 3).[1]

In turn, there has been a considerable and continuous increase in both morbidity and resulting disability-adjusted life years (DALYs) across China. In 2017, the all-age incidence rates and DALYs attributable to dengue fever reached 183.8 and 1.8 per 100,000 respectively, an increase of 5.7 and 4.7 times compared to 1990.[9]

Due to increasing climate suitability, China has also witnessed a notable shift of dengue cases towards northern parts of the country in recent years. In 2019, China experienced an unprecedented multi-site domestic outbreak of dengue in 13 provincial-level administrative divisions (PLADs) which is significantly higher than the average of 3.67 PLADs per year from 2013 to 2018.[2]

Climatic conditions, through the effects of rainfall and temperature on mosquito abundance and dengue transmission rate, are key drivers of dengue fever outbreaks.[10] Research warns that climate change may lead to increased dengue risk in China, especially within its south and southeast coast regions, and other places with no domestic outbreak of dengue previously recorded.[11] Although dengue fever has not been endemic to date, policymakers in these especially vulnerable regions and nationwide need to advance understanding of how climate influences vector-borne disease transmission. The COVID-19 pandemic has made the risks of health-service collapse due to emerging infectious disease threats acutely clear, and it is imperative that such risks are minimised through strengthened surveillance systems to support planning and preparedness.

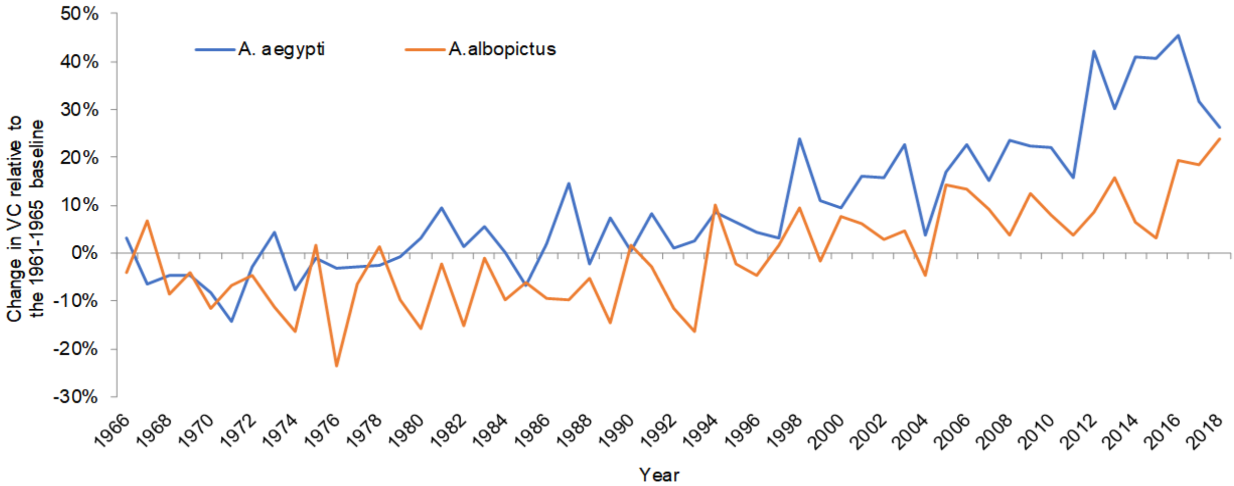

Figure 3: Change in the annual average vectorial capacity (VC) for *A. albopictus* and *A. aegypti* in mainland China, relative to the 1961-1965 baseline

Detection, preparedness, and response to health emergencies

Climate change poses threats to human health by modifying patterns of disease transmission and increasing climate-related extreme events such as heatwaves, floods, cyclones and wildfires. Due to differences in management efficiency, infrastructure construction and social preparation, the ability to detect and quickly respond to health emergencies including climate-sensitive infectious disease outbreaks and extreme events varies between provinces in China. Derived from the "Check-up for China's Cities" initiative, a comprehensive index system was established to measure the capacity for health emergency management at provincial level. The index system consists of 20 indicators of three dimensions: risk exposure and preparedness, detection and response, and resource support and social participation.[2] The index measures the ability to manage public health emergencies, including infectious disease outbreaks, mass illness of unknown origin, serious food and occupational poisoning, as well as the health consequences of climate change. Clear regional differences were found in provinces' ability to manage public health emergencies (Figure 4). While the average index score for health emergencies management across all provinces was 48.1 (out of a possible 100) in 2018, East China reported a higher ability index than other regions, with Jiangsu (69.7), Shandong (68.9) and Beijing (60.9) scoring highest.[2]

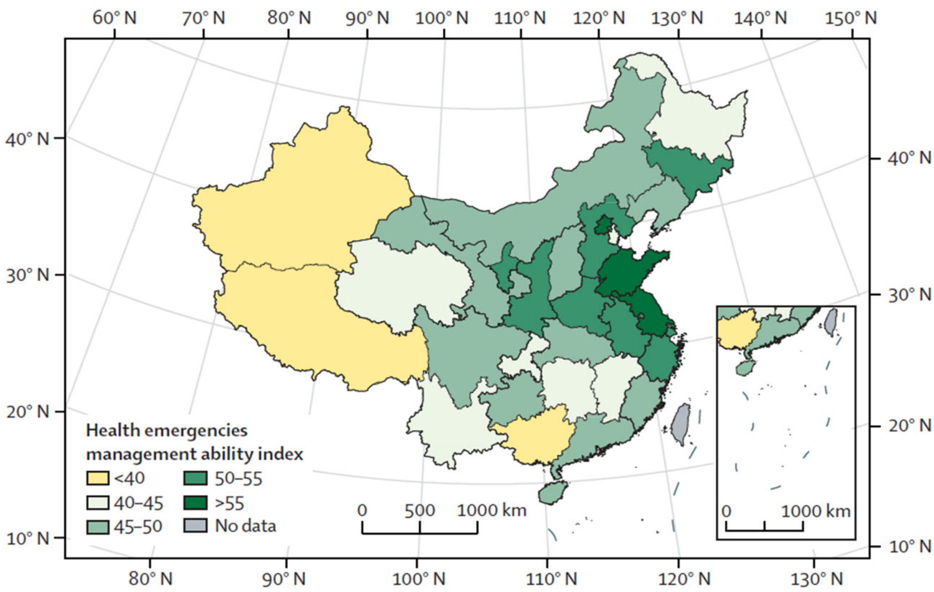

Figure 4: The provincial comprehensive health emergencies management ability index in China

All provinces, even those with the highest scores, should implement measures to strengthen their ability to manage public health emergencies. Shandong performed relatively weakly in the dimension of detection and response, while Beijing has the greatest scope for improvement with regard to risk analysis and planning. The provinces in southwest and northwest China scored relatively low for detection and response, underscoring the crucial need for investment in medical infrastructure. In order to address current and emerging risks to public health, all provinces are encouraged to strengthen their capacities for health emergency management by incorporating measures for prevention, mitigation, preparedness, response and recovery. Special attention to developing proactive approaches to enhancing planned and coordinated rapid-response capability to public health threats will reduce the risk of future health service collapse as experienced during COVID-19.

Investing in a low-carbon economy

China's efforts in coal phase-out and promotion of renewable energy development continue to successfully reshape the national energy mix. The 2020 Chinese Lancet Countdown report tracks energy investments, taking data from Chinese national statistics.[12] Investment in new Chinese coal-fired power generation declined from RMB 197 billion in 2008 to RMB 58 billion in 2019. Investment in renewable energy reached RMB 538 billion in 2019, largely due to investments in solar photovoltaics, as well as increasing investments in wind (Figure 5).[13,14] Correspondingly, the ratio of investment in low-carbon energy (including hydro and nuclear power) to new coal power has risen sharply, from 1:1 in 2008 to 9:1 in 2019. Investment in the overall power grid continues to expand, and reached RMB 508 billion in 2018 and RMB 446 billion in 2019, increasing capacity of the grid for integration of intermittent renewable energy.[2] Driven by shifting investment patterns, overall coal share in total primary energy supply (TPES) has declined, while the national share of low-carbon electricity increased steadily, leading to improved air quality and extensive health benefits.

Despite the substantial progress in coal phaseout and renewable energy development in recent years, coal energy projects have in fact accelerated in 2020 since the onset of the coronavirus outbreak. From January to June alone, China permitted construction of 17.0 GW new coal-fired power- more than the previous two years combined (12.0 GW).[15] Moreover, 57 new coal plant projects (78 GW) progressed toward completion between January and May 2020, as a result of relaxed restrictions on new builds.[16] The surge in new coal-fired power plant approvals is being driven in large part by provincial governments to boost post-pandemic economy. Economic stimulus plans that support new coal capacity would pose a significant risk of carbon lock-in,[17] jeopardise the realisation of the 1.5°C climate target,[18] and exacerbate health threats due to air pollution and climate change.

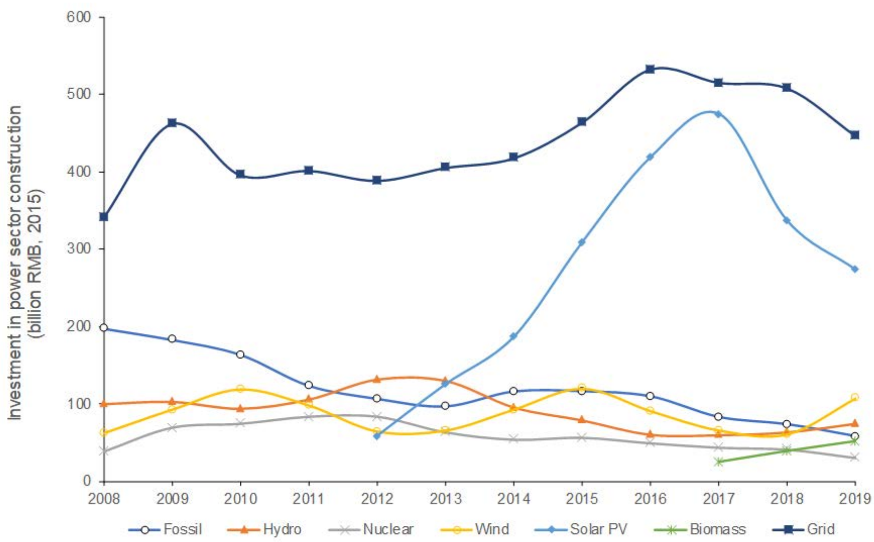

Figure 5: Investment in power sector construction from 2008 to 2019

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Organisations and acknowledgements

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THE LANCET COUNTDOWN

The Lancet Countdown: Tracking Progress on Health and Climate Change is an international, multi-disciplinary collaboration that exists to monitor the links between public health and climate change. It brings together 38 academic institutions and UN agencies from every continent, drawing on the expertise of climate scientists, engineers, economists, political scientists, public health professionals and doctors. Each year, the Lancet Countdown publishes an annual assessment of the state of climate change and human health, seeking to provide decision-makers with access to high-quality evidence-based policy guidance. For the full 2020 assessment, visit www.lancetcountdown.org/2020-report/

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CHINA CDC

The Chinese Center for Disease Control and Prevention (China CDC) is a governmental and national-level technical organization specialized in disease control and prevention and public health. Its mission is to create a safe and healthy environment, maintain social stability, ensure national security and promote the health of people through prevention and control of disease, injury and disability. Devoted to disease control and prevention, it relies on scientific research and values talent as a fundamental element.